Review of genetics terms

A thorough knowledge of the specialist terms used in Genetics is essential for an understanding of the subject. Read this page carefully. Using the information given and a separate text book (or other source) write your own definitions for the terms shown in italics.

The *gene* is the basic unit of inheritance for a given trait. *Alleles* are alternative forms of a gene. For example genes can code for eye colour and the alleles include varying shades of brown, blue, green and even violet.

A *dominant* allele produces a trait that overrides the presence of another trait (*dominate* to control or rule). A capital (upper case) letter indicates dominant traits. For example, brown eyes are a dominant trait over blue eyes. If an individual receives a brown (B) allele from its mother and a brown from its father, the genotype is BB and brown eyes would be present. If the same individual received a brown allele from its mother and a blue from its father, the genotype would be Bb and it would still have brown eyes. The only way for the blue trait to appear is if it is the only one present (bb). This is because the blue allele is *recessive* (meaning to recede or go back). A small (lower case) letter indicates recessive traits.

Homo means 'the same' and *homozygous* allele pairs are the same, such as BB (meaning two dominant, brown alleles) or bb (two recessive blues). *Hetero* means 'different' and *heterozygous* allele pairs contain different alleles, such as Bb.

The *genotype* refers to the complete genetic make up of an individual whether or not it is expressed. The *phenotype* describes what we see (from a Greek version of *pheno,* meaning 'what is seen'). An individual may carry a Bb eye colour genotype. In other words one brown and one blue eye colour allele. However, the phenotype will be brown. The brown eyes do not tell us if the individual is homozygous dominant (BB), meaning that both dominant alleles are present, or heterozygous dominant (Bb), meaning two alleles are present but only the dominant one is being expressed.

As the name indicates, *lethal* alleles can kill. Some lethal alleles need only be present in one dose to kill. In this case they are called *dominant lethal*, while in others a double dose is required (recessive lethal). *Conditional lethals* are alleles that will only be lethal under certain conditions. For example, some will only be activated in higher than normal temperatures. If the organism carrying the conditional lethal is not exposed to the higher temperatures, it remains unharmed. Lethal alleles are not usually found in high frequencies in populations though, as always, exceptions occur.

There are relatively few *chromosomes* in the nucleus of a cell of an organism. Because there are so many inheritable characteristics, each chromosome must consist of many genes. A chromosome consists of a *centromere,* a circular structure, and paired strands called *chromatids* attached to the centromere.

The term *autosome* refers to all chromosomes that are not sex chromosomes. *Sex chromosomes* are those that determine the sex an organism will become. Lethal genes can occur on autosomes or sex chromosomes. The term *autosomal lethal* tells us that the lethal gene is located on a non-sex chromosome. The frequency of *sex-linked lethals* can vary. If the lethal is dominant, only one dose is required to kill so either sex can expect fairly equal ratios.

If the lethal is *X-linked recessive* the distribution will differ between the sexes. This is because females receive two X chromosomes and would therefore require a double dose of the recessive for it to be expressed (therefore lethal) whereas males, who receive one X and one Y, would be affected by a single dose. As the Y chromosome carries very few genes, sex-linked lethals are also called *X-linked* because they are found on the X chromosome.

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